RELIABILITY TESTING QUESTIONNAIRE

Apache JMeter Performance and Reliability Evaluation

Integrated Smart Campus Vehicle Management System (ISCVMS)

Camarines Sur Polytechnic Colleges

Updated from the corrected JMeter runs and the admin record check on 18 October 2026. Figures are Summary Report values.

Part A — Tester and Test Environment Profile

Field	Answer
Tester / Respondent Name	Adrian Adams M. Daduya
Role being tested	Researcher
Date of Testing	18 October 2026
Target URL	https://www.iscvms.com
Apache JMeter Version	5.6.3
Machine Specs (CPU / RAM)	AMD Ryzen 5 7520U with Radeon Graphics / 7.2 GB RAM
Network Condition	Other — home internet to the public HTTPS site
Test Plan Name / File	ISCVMS-Reliability.jmx
Modules Targeted	Login, Registration, Portals, RFID API, Reports

Part B — Apache JMeter Test Results Log

No.	Scenario	Threads	Ramp-up (s)	Loops / duration	Avg (ms)	Min / Max (ms)	Throughput	Error %	Interpretation
1	Login (normal load) — GET /login	10	10	5 loops (50 samples)	175	72 / 861	5.2/sec	0.00%	Stable. Sign In page stayed available.
2	Login (moderate load) — GET /login	30	30	3 loops (90 samples)	148	71 / 781	3.1/sec	0.00%	Stable. Faster on average than the 10-user run.
3	Registration page — GET /register	10	10	3 loops (30 samples)	348	92 / 1571	3.1/sec	0.00%	Registration form stayed available.
4	Admin login and GET /admin	1	1	1	See tree	See tree	See tree	0.00% on the tree	Corrected. Response title is Admin Dashboard.
5	RFID scan — POST /api/rfid/scan	5	10	6	See tree	See tree	See tree	Red in JMeter	Corrected. Body is Card Not Registered for GATE-IN-1. Red rows are the 404 denial, not a crash.
6	RFID heartbeat — POST /api/rfid/heartbeat	10	10	6 loops (60 samples)	104	45 / 1200	6.3/sec	0.00%	Stable. Body is ok true for GATE-IN-1.
7	Mixed portal browsing	5	10	3	See tree	See tree	See tree	Green on the tree	Corrected. Response title is Admin

									Dashboard.
8	Soak / sustained run	10	30	600 seconds (23,189 samples)	251	55 / 2793	38.6/sec	0.00%	Stable for 10 minutes. Visitor pre-register was slowest (avg 502 ms).

Rows 4, 5, and 7 were corrected after the first Summary Reports. Those first totals were the Sign In page and an invalid-token scan, so their old millisecond figures are not used. The corrected proof is the View Results Tree.

Row 8 detail, all 0.00% error: Home avg 106 ms (5,799); Login page 127 ms (5,799); Register page 269 ms (5,797); Visitor pre-register 502 ms (5,794).

Part C — Reliability Test Cases

No.	Test area	Actual result	Status	Remarks
1	Availability — login, normal load	10 users, 50 samples, avg 175 ms, error 0.00%. Sign In HTML returned.	Pass	Success rate 100%.
2	Availability — login, moderate load	30 users, 90 samples, avg 148 ms, error 0.00%. No crash.	Pass	50-user run also stayed at 0.00% error.
3	Reliability — repeated login	Login page stayed at 0.00% error for 10 users, 30 users, 50 users, and 5,799 soak samples.	Pass	Corrected GET /admin now shows the Admin Dashboard.
4	Reliability — registration under concurrent users	GET /register, 10 users, 30 samples, avg 348 ms, error 0.00%. The form loaded.	Pass	An invalid registration POST was not saved as its own Summary Report.
5	Reliability — RFID scan and heartbeat	Corrected scan body: Card Not Registered, granted false, GATE-IN-1. Latest heartbeat: 60 samples, avg 104 ms, error 0.00%, body ok true.	Pass	JMeter paints the scan red because the denial is HTTP 404. The API stayed up.
6	Fault tolerance — invalid passwords	8 GET /login and 8 POST /login, error 0.00%, avg 344 ms. The page said Too many login attempts.	Pass	The lockout returned the login page. It did not take the site down.
7	Fault tolerance — empty RFID body	One POST /api/rfid/scan with an empty uid: 1 sample, avg 336 ms, error 100%. Heartbeat immediately after: 60 samples, error 0.00%, ok true.	Pass	The bad request was rejected and the gate API kept answering.
8	Recoverability — after the heavier run	After 50 users, error was still 0.00% and the average was 197 ms, close to the 175 ms baseline.	Pass	No outage to recover from. Throughput fell from 3.1/sec at 30 users to 1.7/sec at 50 users.
9	Stability — soak	10 users, 600 seconds,	Pass	JMeter completed the

		23,189 samples, avg 251 ms, max 2,793 ms, error 0.00%.		full 10 minutes.
10	Data integrity — records after the load	Admin reviewed Reports, Registrations, one existing user, and System health. Existing records were unchanged.	Pass	The check did what was required. No corrupted or unexpected new student records were found.

Part D — Performance / Stress Indicators

No.	Test area	Actual result	Status	Remarks
1	Login page response, normal load	Average 175 ms, min 72, max 861, error 0.00%, 10 users.	Pass	Under a 3,000 ms campus-use line.
2	Dashboard / portal response	Corrected GET /admin and the mixed portal samples show the title Admin Dashboard.	Pass	The page identity is confirmed. A new Summary Report for that corrected run was not attached, so no new millisecond total is claimed.
3	Throughput, concurrent users	50-user login page: 1.7 requests/sec, 100 samples, error 0.00%. Soak: 38.6 requests/sec, error 0.00%.	Pass	The site kept answering.
4	RFID latency	Latest heartbeat average 104 ms, min 45, max 1,200, error 0.00%, 60 samples.	Pass	Average is under the 350 ms target. One sample reached 1,200 ms.
5	Error rate under higher concurrency	30 users: 0.00%. 50 users: 0.00%. Soak: 0.00% on 23,189 samples. Record check passed.	Pass	No error spike and no corrupted records in the admin check.
6	Stress limit	50 users, 100 samples, avg 197 ms, min 64, max 656, error 0.00%, 1.7/sec. Compared with 30 users at avg 148 ms and 3.1/sec.	Pass	No crash and no error spike. Throughput dropped. A hard failure point was not reached.

Part E — Reliability Evaluation Rating

Scale: 5 Strongly Agree, 4 Agree, 3 Neutral, 2 Disagree, 1 Strongly Disagree, N/A not run.

No.	Statement	Rating	Why this rating
1	Maturity — consistent during repeated runs	5 Strongly Agree	Login stayed at 0.00% error from 10 users through 50 users and the 10-minute soak.
2	Availability — ISCVMS stayed accessible	5 Strongly Agree	Login, registration, the soak pages, and the corrected admin page all came back.
3	Fault tolerance — invalid requests did not take the site	5 Strongly Agree	Wrong passwords produced Too many login attempts. An

	down		empty RFID body was rejected, and the next heartbeat was ok true.
4	Recoverability — acceptable behavior after a heavier load	4 Agree	50 users did not cause errors. The average moved from 148 ms to 197 ms and throughput fell to 1.7/sec.
5	Stability — soak did not become unusable	5 Strongly Agree	23,189 samples, error 0.00%, maximum 2,793 ms.
6	Data integrity — sampled records stayed accurate	5 Strongly Agree	Reports, Registrations, one existing user, and System health were checked after the load. Existing records were unchanged.
7	Gate API reliability	5 Strongly Agree	Unknown card: Card Not Registered. Heartbeat: ok true, average 104 ms.
8	Portal reliability under concurrent load	4 Agree	The corrected mixed-portal tree shows Admin Dashboard. A new Summary Report for that rerun was not attached.
9	Error handling stayed clear and the system stayed stable	5 Strongly Agree	The lockout text and the RFID denial were clear, and later requests still succeeded.
10	Overall reliability for a pilot	5 Strongly Agree	Pages, lockout, the gate API, and the after-test records all support pilot use. The only measured change under 50 users was slower throughput, not failure.

Part F — Open-Ended Questions

Which JMeter scenarios showed the most stable behavior?

The 10-minute soak, the login page from 10 through 50 users, and the RFID heartbeat. The soak completed 23,189 requests with 0.00% error. The 50-user login page completed 100 requests with 0.00% error. The latest heartbeat completed 60 requests with 0.00% error and an average of 104 ms.

At what load level did the system start to degrade, and what failed first?

It did not fail through 50 users. Error stayed 0.00%. What changed first was throughput: 5.2/sec at 10 users, 3.1/sec at 30 users, and 1.7/sec at 50 users. The average moved from 148 ms at 30 users to 197 ms at 50 users. During the soak, visitor pre-registration was the slowest page at an average of 502 ms. The RFID scan rows that look failed in JMeter are HTTP 404 denials for an unregistered card, not a load collapse.

Were any reliability risks found that must be fixed before deployment?

No outage and no corrupted records were found. The wrong-password lockout and the empty RFID body both failed safely. The admin check of Reports, Registrations, one existing user, and System health passed. The login lockout is intentional protection, not a defect.

What hardware or network limits may have affected the JMeter results?

The test PC is an AMD Ryzen 5 7520U with 7.2 GB RAM, on a home internet connection to the public site rather than localhost. JMeter and the browser shared that PC. The live site also limits login posts, registration posts,

RFID scans, and RFID heartbeats per minute. The 50-user test requested only the login page, so it did not trip the login-post limit.

Recommendations to improve ISCVMS reliability and performance

Keep the rate limits. Treat HTTP 404 on an unknown RFID card as a successful denial when reading JMeter. The visitor pre-register page was the slowest sustained page and is the first public page to profile if a faster form is needed. No reliability defect from these tests has to be fixed before a pilot.

Acknowledgment

Apache JMeter 5.6.3 tests were performed on https://www.iscvms.com. The metrics above are the observed results, including the corrected admin, portal, and RFID-scan screens. The after-test admin record check passed.

Tester / Evaluator: Adrian Adams M. Daduya

Date: 18 October 2026

Signature: _____